DO NOW — RECALL

Appendix 5d OCR notation challenge

Name: _____ Class: _____ Date: _____

1. Why can OCR array[5] access index4 but not index5?

2. What is byRef?

3. What does readLine return?

4. In C# array [2,3,5,7], what is a[2] and total sum?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Five items are indexed 0–4.

2. Model answer

Parameter passing allowing modification of caller's underlying variable.

3. Model answer

Next line of text from a file.

4. Model answer

a[2]=5; total=17.

By the end, you can…

• Use exact indexing

• Trace subroutines and parameters

• Read file operations

Vocabulary — know these terms

array — Fixed-length zero-indexed sequence.

byVal — Pass a value/copy.

byRef — Pass a reference to original.

readLine — Read the next file line.

NEW LEARNING

Read, annotate and explain

Use exact indexing

OCR arrays use zero-based indexing; the number in array declaration is the number of elements.

Trace subroutines and parameters

A procedure does not produce a return value; a function returns a result. byVal passes a copy; byRef allows the original parameter to change.

Read file operations

openRead, readLine, endOfFile and close follow the OCR Appendix 5d conventions. Always close an opened file.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace a 3-element OCR array and identify the valid indexes.

Predict first

Array declaration creates three positions: indexes 0,1,2.

Check reasoning

scores[1] holds 4, so output is 4.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace the OCR reference-language array below. State the output and explain why accessing a[3] would be invalid. [5]
CODE:

```
array a[3]
a[0] = 4
a[1] = 8
a[2] = 12
print(a[2])
```

Question 2

Write OCR exam reference language to read and print every line of the file scores.txt, and then close it. [7]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare procedure addTwo(x:byVal) with addTwo(x:byRef), each executing x=x+2, for caller n=5. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

array values[4] can validly access values[4].

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.